# NEET TEST HUB

## NTH-PREMIUM TEST SERIES 12

| Date – 28 December 2025 | Duration – 3 Hours | Marks – 720 |
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**Physics :-** Moving Charges & Magnetism, Magnetism & Matter, EMI, Alternating Current, Electromagnetic Waves

**Chemistry** :- The d & f-Block Elements , Coordination Compounds, The p Block Elements (Group 13 to 18), Salt Analysis

**Biology :-** Principles of Inheritance & Variation, Molecular Basis of Inheritance, Evolution

### Instruction

1. The test is of 3 hours duration.

2. The test booklet consists of 200 questions. The maximum mark is 720.

3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

5. There is only one correct response for each question.

6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

## Physics

1. $AC$ alternating emf $\mathrm{E} = 110\sqrt{2}\sin 100\mathrm{t}$ volt is applied to a capacitor of $2\mu\mathrm{F}$, the rms value of current in the circuit is.. . . . . .$\mathrm{mA}$.

   A) $22$ B) $20$ C) $25$ D) $30$

2. In a series resonant $LCR$ circuit, the voltage across $R$ is $100\ volts$ and $R = 1\ k\Omega$ with $C = 2\ \mu f$ .The resonant frequency $\omega$ is $200\ rad/s$. At resonance the voltage across $L$ is

   A) $40\,v$ B) $250\,v$
   C) $4 \times 10^{-3}\,v$ D) $2.5 \times 10^{-2}\,v$

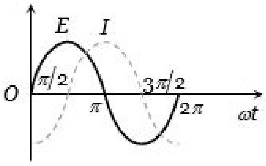
3. In an ac circuit, $V$ and $I$ are given by $V = 150\,sin\,(150t)\ V$ and $I = 150\ \sin\ \left(150t + \frac{\pi}{3}\right) A$. The power dissipated in the circuit is.......$W$

   A) $106$ B) $150$ C) $5625$ D) $0$

4. A $280$ ohm electric bulb is connected to $200V$ electric line. The peak value of current in the bulb will be

   A) About one ampere B) Zero
   C) About two ampere D) About four ampere

5. The variation of the instantaneous current $(I)$ and the instantaneous emf $(E)$ in a circuit is as shown in fig. Which of the following statements is correct




   A) The voltage lags behind the current by $\pi\ /2$
   B) The voltage leads the current by $\pi\ /2$
   C) The voltage and the current are in phase
   D) The voltage leads the current by $\pi$

6. In a circuit, the value of the alternating current is measured by hot wire ammeter as $10$ ampere. Its peak value will be......$A$

   A) $10$ B) $20$ C) $14.14$ D) $7.07$

7. If $E_0$ represents the peak value of the voltage in an ac circuit, the r.m.s. value of the voltage will be

   A) $\frac{E_0}{\pi}$ B) $\frac{E_0}{2}$
   C) $\frac{E_0}{\sqrt{\pi}}$ D) $\frac{E_0}{\sqrt{2}}$

8. The $r.m.s.$ value of an ac of $50Hz$ is $10amp$. The time taken by the alternating current in reaching from zero to maximum value and the peak value of current will be

   A) $2 \times 10^{-2}\ sec$ and $14.14\ amp$
   B) $1 \times 10^{-2}\ sec$ and $7.07\ amp$
   C) $5 \times 10^{-3}\ sec$ and $7.07\ amp$
   D) $5 \times 10^{-3}\ sec$ and $14.14\ amp$

9. The ratio of peak value and r.m.s value of an alternating current is

   A) $1$ B) $\frac{1}{2}$ C) $\sqrt{2}$ D) $1/\sqrt{2}$

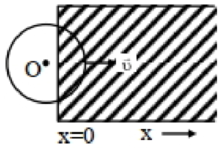
10. The self inductance of a coil having $500$ turns is $50\ mH$. The magnetic flux through the cross-sectional area of the coil while current through it is $8\,mA$ is found to be

    A) $4 \times 10^{-4}\,Wb$ B) $0.04\,Wb$
    C) $4\,\mu Wb$ D) $40\,mWb$

11. A circular loop of radius $0.3\ cm$ lies parallel to a much bigger circular loop of radius $20\ cm$. The centre of the smaller loop is on the axis of the bigger loop. The distance between their centres is $15\ cm$. If a current of $2.0\ A$ flows through the smaller loop, then the flux linked with bigger loop ............ $\times 10^{-11} weber$

    A) $9.1$ B) $6$ C) $3.3$ D) $6.6$

12. A constant magnetic field of $1\,T$ is applied in the $x > 0$ region. A metallic circular ring of radius $1\ m$ is moving with a constant velocity of $1\ m/s$ along the $x$-axis. At $t = 0\ s$, the centre of $O$ of the ring is at $x = -1\,m$. What will be the value of the induced $emf$ in the ring at $t = 1\ s$ ? (Assume the velocity of the ring does not change.) (In $V$)




    A) $1$ B) $4$ C) $2$ D) $0$

13. Consider a circular coil of wire carrying constant current $I$, forming a magnetic dipole. The magnetic flux through an infinite plane that contains the circular coil and excluding the circular coil area is given by $\phi_i$. The magnetic flux through the area of the circular coil area is given by $\phi_0$. Which of the following option is correct?

    A) $\phi_i = -\phi_0$
    B) $\phi_i = \phi_0$
    C) $\phi_i < \phi_0$
    D) $\phi_i > \phi_0$

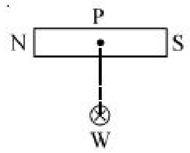
14. The phase difference between the flux linked with a coil rotating in a uniform magnetic field and induced $e.m.f.$ produce in it is

    A) $\pi$ B) $\pi/2$ C) $\pi/3$ D) $-\pi/6$

15. A circular disc of radius $0.2\ meter$ is placed in a uniform magnetic field of induction $\frac{1}{\pi}\ Wb/m^2$ in such a way that its axis makes an angle of $60^o$ with $\vec{B}$. The magnetic flux linked with the disc is.....$Wb$

    A) $0.08$ B) $0.01$ C) $0.02$ D) $0.06$

16. Figure shows a bar magnet and a long straight wire $W$, carrying current into the plane of paper. Point $P$ is the point of intersection of axis of magnet and the line of shortest distance between magnet and the wire. If $P$ is the midpoint of the magnet, then which of the following statements is correct ?




A) magnet experiences a torque in clockwise direction
B) magnet experiences a torque in anticlockwise direction
C) magnet experiences a force, normal to the line of shortest distance
D) magnet experiences a force along the line of shortest distance

17. A circular loop of radius $R$ carrying current $I$ lies in $x-y$ plane with its centre at origin.The total magnetic flux through $x-y$ plane is

A) Directly proportional to $I$
B) Directly proportional to $R$
C) Directly proportional to $R^2$
D) Zero

18. The unit of magnetic flux is

A) $Weber/m^2$ B) Weber C) Henry D) Ampere/m

19. Due to a small magnet intensity at a distance $x$ in the end on position is $9\ Gauss$. What will be the intensity at a distance $\frac{x}{2}$ on broad side on position..... $Gauss$

A) $9$ B) $4$
C) $36$ D) $4.5$

20. Dip angle in vertical plane at an angle $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$ from magnetic meridian is $60^o$, then actual dip at that place

A) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$
B) $\tan^{-1}\left(\frac{1}{\sqrt{6}}\right)$
C) $\tan^{-1}(1)$
D) $\tan^{-1}\left(\sqrt{\frac{3}{2}}\right)$

21. Which statement is correct

A) Magnetic field of current element is perpendicular to position vector
B) Electric field of point charge is along position vector
C) Magnetic mono pole physically does not exist
D) All of above

22. Due to a small magnet, intensity at a distance $x$ in the end on position is $9\ gauss$ . What will be the intensity at a distance $\frac{x}{2}$ on broad side on position....$gauss$

A) $9$ B) $4$ C) $36$ D) $4.5$

23. A ferromagnetic material is heated above its curie temperature. Which one is a correct statement

A) Ferromagnetic domains are perfectly arranged
B) Ferromagnetic domains becomes random
C) Ferromagnetic domains are not influenced
D) Ferromagnetic material changes itself into diamagnetic material

24. Tangent galvanometer is used to measure

A) Charge B) Angle
C) Current D) Magnetic intensity

25. A thin rectangular magnet suspended freely has a period of oscillation equal to $T$. Now it is broken into two equal halves (each having half of the original length) and one piece is made to oscillate freely in the same field. If its period of oscillation is $T'$, then ratio $\frac{T'}{T}$ is

A) $\frac{1}{4}$ B) $\frac{1}{2\sqrt{2}}$ C) $\frac{1}{2}$ D) $2$

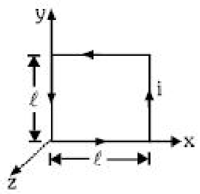
26. Two lines of force due to a bar magnet

A) Intersect at the neutral point
B) Intersect near the poles of the magnet
C) Intersect on the equatorial axis of the magnet
D) Do not intersect at all

27. Magnetic lines of force due to a bar magnet do not intersect because

A) A point always has a single net magnetic field
B) The lines have similar charges and so repel each other
C) The lines always diverge from a single point
D) The lines need magnetic lenses to be made to intersect

28. The magnetic field existing in a region is given by $\vec{B} = B_0\ (5+\frac{x}{l})\ \hat{K}$ A square loop of edge $l$ and carrying a current $i$ is placed with its edges parallel to $x-y$ axes. Find the magnitude of the net magnetic force experienced by the loop




A) $5\,B_0\,il$ B) $6\,B_0\,il$
C) $B_0\,il$ D) Zero

29. A charge particle of charge $q$ and mass $m$ is accelerated through a potential diff. $V\ volts$. It enters a region of orthogonal magnetic field $B$. Then radius of its circular path will be

A) $\sqrt{\frac{Vm}{2qB^2}}$
B) $\frac{2Vm}{qB^2}$
C) $\sqrt{\frac{2Vm}{q}}\left(\frac{1}{B}\right)$
D) $\sqrt{\frac{Vm}{q}}\left(\frac{1}{B}\right)$

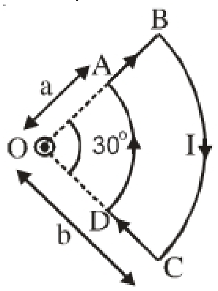
30. The pole strength of a bar magnet is $48\ ampere-metre$ and the distance between its poles is $25\ cm$ . The moment of the couple by which it can be placed at an angle of $30°$ with the uniform magnetic intensity of flux density $0.15\ Newton/ampere-metre$ will be.......$Newton \times metre$

A) $12$ B) $18$
C) $0.9$ D) None of the above

**31.** A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected in the region such that its velocity is pointed along the direction of fields, then the electron

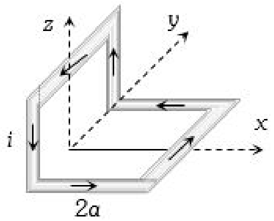
**A)** will turn towards right of direction of motion
**B)** will turn towards left of direction of motion
**C)** speed will increase
**D)** speed will decrease

**32.** A current loop $ABCD$ is held fixed on the plane of the paper as shown in the figure. The arcs $BC$ (radius $= b$) and $DA$ (radius $= a$) of the loop are joined by two straight wires $AB$ and $CD$. A steady current $I$ is flowing in the loop. Angle made by $AB$ and $CD$ at the origin $O$ is $30^o$. Another straight thin wire with steady current $I_1$ flowing out of the plane of the paper is kept at the origin.
The magnitude of the magnetic field $(B)$ due to the loop $ABCD$ at the origin $(O)$ is :




**A)** $0$
**B)** $\frac{\mu_0 I(b-a)}{24ab}$
**C)** $\frac{\mu_0 I}{4\pi}\left[\frac{b-a}{ab}\right]$
**D)** $\frac{\mu_0 I}{4\pi}\left[2(b-a)+\frac{\pi(a+b)}{3}\right]$

**33.** A non-planar loop of conducting wire carrying a current $I$ is placed as shown in the figure. Each of the straight sections of the loop is of length $2a$. The magnetic field due to this loop at the point $P\,(a,0,a)$ points in the direction




**A)** $\frac{1}{\sqrt{2}}(-\hat{j}+\hat{k})$
**B)** $\frac{1}{\sqrt{3}}(-\hat{j}+\hat{k}+\hat{i})$
**C)** $\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$
**D)** $\frac{1}{\sqrt{2}}(\hat{i}+\hat{k})$

**34.** A current of $I\ ampere$ is passed through a straight wire of length $2.0\ metres$. The magnetic field at a point in air at a distance of $3\ metres$ from either end of wire and lying on the axis of wire will be

**A)** $\frac{\mu_0}{2\pi}$ **B)** $\frac{\mu_0}{4\pi}$
**C)** $\frac{\mu_0}{8\pi}$ **D)** Zero

**35.** When a certain length of wire is turned into one circular loop, the magnetic induction at the centre of coil due to some current flowing is $B_1$ If the same wire is turned into three loops to make a circular coil, the magnetic induction at the center of this coil for the same current will be

**A)** $B_1$ **B)** $9\,B_1$
**C)** $3\,B_1$ **D)** $27\,B_1$

**36.** A milliammeter of range $10\ mA$ has a coil of resistance $1\ \Omega$. To use it as voltmeter of range $10\ volt$, the resistance that must be connected in series with it, will be ............ $\Omega$

**A)** $999$ **B)** $99$ **C)** $1000$ **D)** None of these

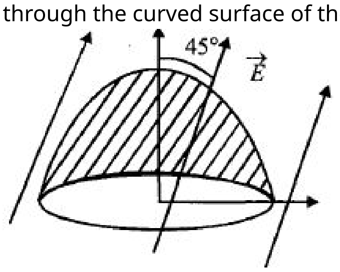
**37.** When a conducting soap bubble is negatively charged then

**A)** Its size starts varying arbitrarily
**B)** It expands
**C)** It contracts
**D)** No change in its size takes place

**38.** $A$ and $B$ are two points on the axis and the perpendicular bisector respectively of an electric dipole. $A$ and $B$ are far away from the dipole and at equal distance from it. The field at $A$ and $B$ are $\vec{E}_A$ and $\vec{E}_B$.

**A)** $\vec{E}_A = \vec{E}_B$
**B)** $\vec{E}_A = 2\vec{E}_B$
**C)** $\vec{E}_A = -2\ \vec{E}_B$
**D)** $|E_B| = \frac{1}{2}|E_A|$ , and $\vec{E}_B$ is perpendicular to $\vec{E}_A$

**39.** The flat base of a hemisphere of radius $a$ with no charge inside it lies in a horizontal plane. A uniform electric field $\vec{E}$ is applied at an angle $\frac{\pi}{4}$ with the vertical direction. The electric flux through the curved surface of the hemisphere is




**A)** $\pi a^2 E$
**B)** $\frac{\pi a^2 E}{\sqrt{2}}$
**C)** $\frac{\pi a^2 E}{2\sqrt{2}}$
**D)** $\frac{(\pi+2)\ \pi a^2 E}{(2\sqrt{2})^2}$

**40.** Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of $30^\circ$ with each other. When suspended in a liquid of density $0.8\ g\,cm^{-3}$, the angle remains the same. If density of the material of the sphere is $1.6\ g\,cm^{-3}$, the dielectric constant of the liquid is

**A)** $2$ **B)** $1$ **C)** $4$ **D)** $3$

**41.** Two spherical conductors $B$ and $C$ having equal radii and carrying equal charges in them repel each other with a force $F$ when kept apart at some distance. A third spherical conductor having same radius as that of $B$ but uncharged is brought in contact with $B$, then brought in contact with $C$ and finally removed away from both. The new force of repulsion between $B$ and $C$ is

**A)** $F/4$ **B)** $3F/4$ **C)** $F/8$ **D)** $3F/8$

**42.** Charge on $\alpha$ - particle is

**A)** $4.8\times10^{-19}\,C$ **B)** $1.6\times10^{-19}\,C$

C) $3.2 \times 10^{-19} C$  D) $6.4 \times 10^{-19} C$

**43.** A body has $-80$ $micro\ coulomb$ of charge. Number of additional electrons in it will be

A) $8 \times 10^{-5}$  B) $80 \times 10^{-17}$
C) $5 \times 10^{14}$  D) $1.28 \times 10^{-17}$

**44.** What is the magnitude of a point charge due to which the electric field $30$ $cm$ away has the magnitude $2$ $newton/coulomb$ $[1/4\pi\varepsilon_0 = 9 \times 10^9$ $Nm^2/C^2]$

A) $2 \times 10^{-11}$ $coulomb$  B) $3 \times 10^{-11}$ $coulomb$
C) $5 \times 10^{-11}$ $coulomb$  D) $9 \times 10^{-11}$ $coulomb$

**45.** A body can be negatively charged by

A) Giving excess of electrons to it
B) Removing some electrons from it
C) Giving some protons to it
D) चालक पर आवेश एवं उसके क्षेत्रफल दोनों के व्युत्क्रमानुपाती होता है

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## Chemistry

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**46.** Aqueous solution of $Ni^{2+}$ contains $[Ni(H_2O)_6]^{2+}$ and its magnetic moment is 2.83 BM. When ammonia is added in it, comment on the magnetic moment of solution

A) It will remain same
B) It increases from 2.83 BM
C) It decreases from 2.83 BM
D) It cannot be predicted theoretically

**47.** Which of the following ligand gives chelate complexes?

A) $SCN^-$  B) $C_2O_4^{2-}$  C) Pyridine  D) $NH_2 - \overset{+}{N}H_3$

**48.** What is the coordination number of Ni in nickel DMG complex?

A) 2  B) 3  C) 6  D) 4

**49.** Which of the following complex compound is "Pentaaquacyanidoiron (III) trichloridotricyanido cobaltate (III)"?

A) $[Fe\,(CN)\,(H_2O)_5]\,[CoCl_3(CN)_3]$
B) $[Fe\,(CN)\,(H_2O)_5]\,[CoCl_3(CN)_3]_3$
C) $[Fe(CN)_2(H_2O)_4]_3[FeCl_3(CN)_3]_2$
D) $[Fe\,(CN)\,(H_2O)_5]_3[CoCl_3(CN)_3]_2$

**50.** $KCl.MgCl_2.6H_2O$ is a

A) mixed salt  B) double salt
C) basic salt  D) complex salt

**51.** Select the compound having maximum conductivity in aqueous medium.

A) $[Cr(NH_3)_6]\,Cl_3$  B) $[Cr(NH_3)_5Cl]\,Cl_2$
C) $[Cr(NH_3)_4Cl_2]\,Cl$  D) $[Cr(NH_3)_3Cl_3]$

**52.** Ammonia forms the complex ion $[Cu(NH_3)_4]^{2+}$ with copper ions in the alkaline solutions but not in acidic solutions. What is the reason for it?

A) In acidic solutions, protons coordinate with ammonia molecules forming $NH_4^+$ ions and $NH_3$ molecules are not available
B) In alkaline solutions insoluble $Cu(OH)_2$ is precipitated which is soluble in excess of any alkali
C) Copper hydroxide is an amphoteric substance
D) In acidic solutions hydration protects copper ions

**53.** Which one of the following is an inner orbital complex as well as diamagnetic in behaviour?

A) $[Zn(NH_3)_6]^{2+}$  B) $[Ni(NH_3)_6]^{2+}$
C) $[Cr(NH_3)_6]^{3+}$  D) $[Co(NH_3)_6]^{3+}$

**54.** Which one of the following cyano complexes would exhibit the lowest value of paramagnetic behaviour?

(Atomic no. Cr=24, Mn=25, Fe=26, Co=27)

A) $[Co(CN)_6]^{3-}$  B) $[Fe(CN)_6]^{3-}$  C) $[Mn(CN)_6]^{3-}$  D) $[Cr(CN)_6]^{3-}$

**55.** Among the following, which are ambidentate ligands?
(i) $SCN^-$
(ii) $NO_3^-$
(iii) $NO_2^-$
(iv) $C_2O_4^{2-}$

A) (i) and (iii)  B) (i) and (iv)
C) (ii) and (iii)  D) (ii) and (iv)

**56.** Which of the following statement is not correct?

A) $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ have the same magnetic moment
B) $[NiCl_4]^{2-}$ and $[PtCl_4]^{2-}$ have different shapes
C) Hybridisation state of Co in $[Co(Ox)_3]^{3-}$ is $sp^3d^2$
D) In brown-ring complex $[Fe(H_2O)_5NO]\,SO_4$ oxidation state of Fe is +1

**57.**
Which of the following properties don't help in differentiating, different hydrated isomers of $CrCl_3\,.6H_2O$?

A) Conductivity measurements
B) Precipitation by $AgNO_3$
C) Dipole moment
D) Magnetic moment

**58.** Which isomerism shown by the compound $[Cr\,(NH_3)_5NO_2]\,Cl_2$

A) Co-ordination  B) Geometrical
C) Optical  D) Linkage

**59.** Which of the following species must have maximum number of electrons in '$d_{xy}$' orbital?

A) Cr  B) $Fe^{3+}$  C) $Cu^+$  D) Both A and B

**60.** In an octahedral structure, the pair of d-orbitals involved in $d^2\ sp^3$ hybridisation is

A) $d_{x^2-y^2}, d_{z^2}$  B) $d_{xz}\ , d_{x^2-y^2}$
C) $d_{z^2}, d_{xz}$  D) $d_{xy}, d_{yz}$

**61.** The crystals of ferrous sulphate on heating give

A) $FeO + SO_2 + H_2O$  B) $Fe_2O_3 + H_2SO_4 + H_2O$
C) $Fe_2O_3 + SO_2 + SO_3$  D) $FeO + SO_3 + H_2SO_4 + H_2O$

**62.** Which statements are incorrect about transition elements?

A) They show variable valency
B) They readily form complex compounds
C) All their ions are colourless
D) Their ions contain partially filled d-orbital

**63.** $KMnO_4$ can be prepared from $K_2MnO_4$ as per the reaction
$3MnO_4^{2-} + 2H_2O \rightleftharpoons 2MnO_4^- + MnO_2 + 4OH^-$
The reaction can go to completion by removing $OH^-$ ions by adding

**A)** KOH **B)** $CO_2$ **C)** $SO_2$ **D)** HCl

**64.** The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence:

**A)** $Ga < In < Al < Tl$
**B)** $Al < Ga < In < Tl$
**C)** $Tl < In < Ga < A$
**D)** $In < Tl < Ga < Al$

**65.** The most basic hydroxide from the following is

**A)** $Pr(OH)_3\,(Z = 59)$ **B)** $Sm(OH)_3\,(Z = 62)$
**C)** $Ho(OH)_3\,(Z = 67)$ **D)** $La(OH)_3\,(Z = 57)$

**66.** The most convenient method to protect the bottom of ship made of iron is

**A)** coating it with red lead oxide
**B)** white tin plating
**C)** connecting it with Mg block
**D)** connecting it with Pb block

**67.** When to a copper sulphate solution, excess of ammonium hydroxide added then

**A)** No change is observed
**B)** Blue precipitate of copper hydroxide is obtained
**C)** A deep blue solution is obtained
**D)** Black precipitate of copper oxide is obtained

**68.** Which of the following statements is not correct about the electronic configuration of gaseous chromium atom

**A)** It has 5 electrons in 3d and one electron in 4s orbitals
**B)** The principal quantum numbers of its valence electrons are 3 and 4
**C)** It has 6 electrons in 3d orbital
**D)** Its valance electrons have quantum number $l = 0$

**69.** Which of the following ions will liberate iodine when treated with KI?

**A)** $Cu^{2+}$ **B)** $Fe^{2+}$ **C)** $Pb^{2+}$ **D)** $Sn^{2+}$

**70.** $4AgNO_3 + \underset{(dry)}{2Cl_2} \rightarrow 4X + 2Y + Z$

In the above sequence of reaction y and z are respectively

**A)** $NO_2$ and $N_2O_4$ **B)** $N_2O_5, O_2$
**C)** AgCl and $O_2$ **D)** AgCl, $N_2$

**71.** Among the following transition elements, which one has the highest second ionisation energy?

(i) V $(Z = 23)$ (ii) Cr $(Z = 24)$ (iii) Mn $(Z = 25)$ (iv) Ti $(Z = 22)$

**A)** (i) **B)** (ii) **C)** (iii) **D)** (iv)

**72.** Which one of the following has strongest metallic bonding?

**A)** Fe **B)** Sc **C)** V **D)** Cr

**73.** Increasing basic properties of $TiO_2, ZrO_2$ and $HfO_2$ are in order:

**A)** $TiO_2 < ZrO_2 < HfO_2$
**B)** $HfO_2 < ZrO_2 < TiO_2$
**C)** $HfO_2 < TiO_2 < ZrO_2$
**D)** $ZrO_2 < TiO_2 < HfO_2$

**74.** Lanthanide contraction is caused due to

**A)** The imperfect shielding on outer electrons by 4 f -electrons from the nuclear charge
**B)** The appreciable shielding on outer electrons by 4 f -electrons from the nuclear charge
**C)** The appreciable shielding on outer electrons by 5d-electrons from nuclear charge
**D)** The same effective nuclear charge from Ce to Lu

**75.** Aqua-regia reacts with Pt to yield –

**A)** $Pt(NO_3)_4$ **B)** $H_2PtCl_6$ **C)** $PtCl_4$ **D)** $PtCl_2$

**76.**

$$Mg_3B_2 \xrightarrow{aq.\ HCl} x + MgCl_2$$
$$x \xrightarrow{H_2O} y + H_2$$

for (X) and (Y) the incorrect choice is?

**A)** (X) is $BCl_3$
**B)** (Y) is $H_3BO_3$
**C)** (X) with air and (Y) on strong heating (red heat) give same compound
**D)** In (Y) boron completes is octet by moving $H^+$ from water molecule

**77.**

Identify the correct statements about borazene, $B_3N_3H_6$,:
(i) Borazene is aromatic.
(ii) There are four isomers of bi substituted molecule of borazene molecules ,$(B_3N_3H_4X_2)$.
(iii) Borazene is more reactive towards addition reactions than benzene.

**A)** Only (i)
**B)** (i) and (ii)
**C)** (i) and (iii)
**D)** (i), (ii) and (iii)

**78.** The aqueous solution of $AlCl_3$ is acidic due to the hydrolysis of

**A)** aluminium ion
**B)** chloride ion
**C)** both aluminium and chloride ions
**D)** none of the above

**79.** The reaction of $H_3N_3B_3Cl_3$ (A) with $LiBH_4$ in tetrahydrofuran gives inorganic benzene (B). Further, the reaction of (A) with (C) leads to $H_3N_3B_3(Me)_3$ . Compounds (B) and (C) respectively, are

**A)** Borazine and MeBr
**B)** Boron nitride and MeBr
**C)** Diborane and MeMgBr
**D)** Borazine and MeMgBr

**80.** B has a smaller first ionization enthalpy than Be. Consider the following statements
(i) it is easier to remove 2p electron than 2s electron
(ii) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be
(iii) 2s electron has more penetration power than 2p electron
(iv) atomic radius of B is more than Be
(atomic number B = 5, Be = 4)
The correct statements are

**A)** (i), (ii) and (iv)
**B)** (i), (iii) and (iv)
**C)** (i), (ii) and (iii)
**D)** (ii), (iii) and (iv)

**81.** Which of the following structure is simillar to graphite?

**A)** B **B)** $B_4C$ **C)** $B_2H_6$ **D)** BN

**82.** Which statements is not correct about $H_3BO_3$?

**A)** It is a strong tribasic acid
**B)** It has a layer structure in which planar $BO_3$ units are joined by hydrogen bonds
**C)** It is prepared by acidifying an aqueous solution of borax
**D)** It does not act as proton donor but acts as a Lewis acid by accepting hydroxyl ion

**83.** In the aluminothermite process, aluminium acts as

**A)** an oxidizing agent **B)** a flux
**C)** a reducing agent **D)** a solder

**84.** When orthoboric acid is heated strongly, it gives

**A)** $B_2O_3$ **B)** $H_2B_4O_7$ **C)** $HBO_2$ **D)** B

**85.** During Hoope's process of electrolytic refining of Al, the middle layer is of

**A)** Pure Al **B)** Impure Al
**C)** Cryolite + $BaF_2$ **D)** Alloys of Al, Ca, Si

**86.** $B_2H_6 + NH_3 \rightarrow \text{Addition compound (X)} \xrightarrow{450\text{ K}} Y + Z\,(g)$
In the above sequence Y and Z are respectively

**A)** Borazine, $H_2$ **B)** Boron, $H_2$
**C)** Boron nitride, $H_2$ **D)** Boron, Hydrazine

**87.** Which of the following statement is correct about CO?

**A)** It reduces aqueous solution of $PdCl_2$ to metallic Pd
**B)** CO is neutral oxide and acts as a fuel
**C)** In laboratory it is prepared by dehydrating HCOOH with conc. $H_2SO_4$
**D)** All are correct

**88.** The soldiers of Napoleon army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White metallic tin buttons got converted to grey powder. This transformation is related to

**A)** an interaction with nitrogen of the air at very low temperatures
**B)** a change in the crystalline structure of tin
**C)** an interaction with water vapour contained in the humid air
**D)** a change in the partial pressure of oxygen in the air.

**89.** Aluminium chloride exists as dimmer, $Al_2Cl_6$ in solid state as well as in solution: of non–polar solvents such as benzene. when dissolved in water, it gives

**A)** $Al_2O_3 + 6HCl$ **B)** $Al^{3+} + 3Cl^-$
**C)** $[Al(OH)_6]^{3-} + 3HCl$ **D)** $[Al(H_2O)_6]^{+3} + 3Cl^-$

**90.** Aqueous ammonia is used as a precipitating reagent for $Al^{3+}$ ions as $Al(OH)_3$ rather than aqueous NaOH, because

**A)** $NH_4^+$ is a weak base
**B)** NaOH is a very strong base
**C)** NaOH forms $[Al(OH)_4]^-$ ions
**D)** NaOH forms $[Al(OH)_2]^+$ ions.

## Biology - ( Zoology )

**91.** Who proposed that the genetic code for amino acids should be made up of three nucleotides?

**A)** George Gamow **B)** Francis Crick
**C)** Jacque Monod **D)** Franklin Stahl

**92.** Given below are two statements :

Statement I : Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II : RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.
In the light of the above statements, choose the most appropriate answer from the options given below :

**A)** Both statement I and statement II are correct
**B)** Both statement I and statement II are incorrect
**C)** Statement I is correct but statement II is incorrect
**D)** Statement I is incorrect but statement II is correct

**93.** Match List I with List II:

| | List - I | | List - II |
|---|---|---|---|
| A. | Alfred Hershey and Martha Chase | (i) | Streptococcus pneumo |
| B. | Euchromatin | (ii) | Densely packed and da |
| C. | Frederick Griffith | (iii) | Loosely packed and lig |
| D. | Heterochromatin | (iv) | DNA as genetic mater |

Choose the correct answer from the options given below:

**A)** A-II, B-IV, C-I, D-III
**B)** A-IV, B-II, C-I, D-III
**C)** A-IV, B-III, C-I, D-II
**D)** A-III, B-II, C-IV, D-I

**94.** Given below are two statements :

Statement I : In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.

Statement II: DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.

In the light of the above statements, choose the most appropriate answer from the options given below :

**A)** Both statement I and statement II are correct
**B)** Both statement I and statement II are incorrect
**C)** Statement I is correct but statement II is incorrect
**D)** Statement I is incorrect but statement II is correct

**95.** Which of the following are the post-transcriptional events in an eukaryotic cell?
A. Transport of pre-mRNA to cytoplasm prior to splicing.
B. Removal of introns and joining of exons.
C. Addition of methyl group at 5' end of hnRNA.
D. Addition of adenine residues at 3 ' end of hnRNA.
E. Base pairing of two complementary RNAs.

Choose the correct answer from the options given below :

**A)** A, B, C only **B)** B, C, D only
**C)** B, C, E only **D)** C, D, E only

**96.** Which factor is important for termination of transcription?

**A)** $\alpha$ (alpha) **B)** $\sigma$ (sigma)
**C)** $\rho$ (rho) **D)** $\gamma$ (gamma)

**97.** Twins are born to a family that lives next door to you. The twins are a boy and a girl. Which of the following must be true?

**A)** They are monozygotic twins.
**B)** They are fraternal twins.
**C)** They were conceived through in vitro fertilization.
**D)** They have $75\%$ identical genetic content.

**98.** Which chromosome in the human genome has the highest number of genes?

**A)** Chromosome X **B)** Chromosome $Y$
**C)** Chromosome 1 **D)** Chromosome 10

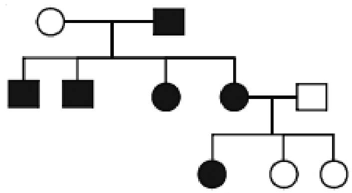
**99.** Histones are enriched with -

**A)** Lysine & Arginine
**B)** Leucine & Lysine
**C)** Phenylalanine & Leucine
**D)** Phenylalanine & Arginine

**100.** Which of the following statements are correct about Klinefelter's Syndrome?
$A$. This disorder was first described by Langdon Down $(1866)$.
$B$. Such an individual has overall masculine development. However, the feminine developement is also expressed.
$C$. The affected individual is short statured.
$D$. Physical, psychomotor and mental development is retarded.
$E$. Such individuals are sterile.
Choose the correct answer from the options given below:

**A)** $A$ and $E$ only
**B)** $A$ and $B$ only
**C)** $C$ and $D$ only
**D)** $B$ and $E$ only

**101.** In the given pedigree, indicate whether the shaded symbols indicate dominant or recessive allele.



**A)** Recessive
**B)** Codominant
**C)** Dominant
**D)** It can be recessive or dominant both

**102.** Sickle cell-anaemia disorder arises due to

**A)** Duplication of a segment of $DNA$
**B)** Substitution in a single base of $DNA$
**C)** Deletion of a segment of $DNA$
**D)** Duplication in a base pair of $RNA$

**103.** Down's syndrome is an example of

**A)** Anueploidy **B)** Polyteny
**C)** Polyploidy **D)** Monoploidy

**104.** Which of these is not a Mendelian disorder?

**A)** Cystic fibrosis **B)** Sickle-cell anaemia
**C)** Colourblindness **D)** Turner's syndrome

**105.** If a man who is colour blind marries a women, who is pure normal for colour vision, the chances of their sons have colour blindness is

**A)** $100\%$ **B)** $50:50$
**C)** $0\%$ **D)** $75:25$

**106.** Dihybrid ratio of test cross $1:1:1:1$ proves that

**A)** $F_1$ hybrid produces four different progenies
**B)** $F_1$ hybrid produces two different progenies
**C)** Parents produce two different progenies
**D)** None of the above

**107.** Mendel obtained recessive character in $F_2$ by ... $A$... the ...$B$... plants. Here $A$ and $B$ refers to

**A)** $A-$self-pollinating; $B-F_1$
**B)** $A-$self-pollinating; $B-F_2$
**C)** $A-$cross-pollinating ; $B-F_1$
**D)** $A-$cross-pollinating; $B-F_2$

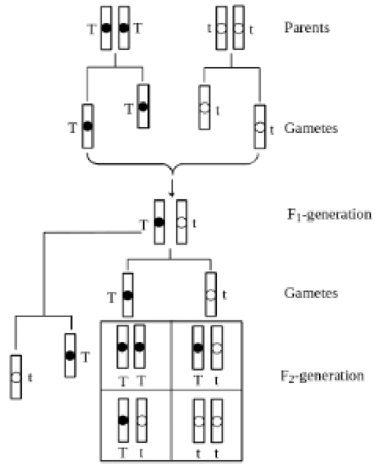
**108.** The number of contrasting characters studied by Mendel for his experiments was

**A)** $7$ **B)** $14$ **C)** $4$ **D)** $2$

**109.** Mendel conducted experiments for

**A)** $7\,years$ **B)** $6\,years$
**C)** $5\,years$ **D)** $4\,years$

110. What does this diagram indicate?




A) Law of dominance interpretated on basis of genes
B) Law of segregation interpretated on basis of genes
C) Law of independent assortment interpretated on basis of genes
D) Simply gamete genes

111. A woman has a haemophilic son and three normal children. Her genotype and that of her husband with respect to this gene would be

A) $XX \wedge X^hY$
B) $X^hX^h \wedge X^hY$
C) $X^hX^h$ and $XY$
D) $X^hX \wedge XY$

112. Which of the following law was discovered first by Mendel?

A) Law of dominance
B) Law of segregation
C) Law of independent assortment
D) Law of sex determination

113. $A$ : Blood group phenotype is controlled by presence or absence of antigens present on surface coating of $RBC$.
$R$ : These antigens are of three types and found in the oligosaccharides rich head regions of a glycophorin

A) Assertion and Reason both are correct and also correct explanation.
B) Assertion and Reason both are correct but not explanation of assertion.
C) Assertion is correct, but Reason is incorrect.
D) Both Assertion and Reason are incorrect.

114. $A$ : Dominance is not an autonomous feature of a gene.
$R$ : It depends as much on the gene product and the production of a particular phenotype from this product

A) Assertion and Reason both are correct and also correct explanation.
B) Assertion and Reason both are correct but not explanation of assertion.
C) Assertion is correct, but Reason is incorrect.
D) Both Assertion and Reason are incorrect.

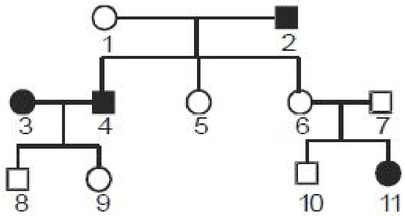
115. While solving the problem of sex determination in large number of insects, it was observed that

A) All eggs lack sex chromosome
B) Some of the sperms bear the $X-$chromosome
C) All eggs as well as sperms bear the $X-$chromosome
D) Some of the eggs bear the $X-$chromosome

116. Which of the following gametes are produced by individual with genotype $RrYy$?

A) $Ry, RY$
B) $ry, ry$
C) $Ry, ry, rY$
D) $Ry, RY, rY, ry$

117. In Huntington's disease, the unaffected persons are homozygous for normal allele $h$. The following is erroneous because




A) it shows both male and female affected by Huntingtons disease
B) either person $6$ or $7$ should have the disease, if individual $11$ shows the disease.
C) at least one of the $2$ children $(8, 9)$ should have the disease
D) all of these

118. Loss of a single chromosome produces a condition called.

A) Haploidy B) Nullisomy C) Trisomy D) Monosomy

119. Which is incorrect in Mendelian characters ?

| Character | dominant | recessive |
|---|---|---|
| A) Pod colour | green | yellow |
| B) Seed shape | round | wrinkled |
| C) Flower position | terminal | Axillary |
| D) Flower colour | violet | white |

120. In a dihybrid cross between $RRYY$ and $rryy$, the number of $RrYy$ $F_2$ genotypes will be

A) $4$ B) $3$ C) $2$ D) $9$

121. In his classic experiments on pea plants, Mendel did not use

A) seed shape
B) flower position
C) seed colour
D) pod length.

122. Between persons of which two blood groups is the blood transfusion not possible

A) $O$ and $AB$ ($AB$ recipient)
B) $O$ and $A$ ($O$ donor)
C) $O$ and $B$ ($O$ donor)
D) $O$ and $AB$ ($AB$ donor)

123. Gene can be defined as

A) Unit of segregation
B) Unit of physiological activity
C) Unit of recombination
D) Unit of function

124. $RR$ $(Red)$ is crossed with $ww$ $(white)$, all the $Rw$ offsprings are pink. This is an indication that $R$ gene is

A) Hybrid
B) Recessive
C) Incompletely dominant
D) Mutant

125. The map distance between genes $A$ and $B$ is $3$ units, between $B$ and $C$ $10$ units and between $C$ and $A$ $7$ units. The order of the genes in a linkage map constructed on the above data would perhaps be

A) $A, B, C$
B) $A, C, B$
C) $B, C, A$
D) $B, A, C$

126. In case of disputed parentage, the blood group analysis of the mother, child and alleged father can

A) Definitely prove a man to be the father
B) Only prove that he cannot be the father
C) Not be of any use
D) None of the above

127. Which one is the sex linked trait

A) Myxoedema
B) Diabetes
C) Blindness
D) Haemophilia

128. Mutation caused by a mutagen is

A) Natural
B) Chemical
C) Spontaneous
D) Induced

129. Pod character i.e. green colour in pea is

A) Dominant
B) Incompletely dominant
C) Recessive character
D) Abnormal character

130. Which of the following has been used for genetic researches

A) Pisum
B) Neurospora
C) E. coli
D) All the above

131. If in a garden pea plant, a cross is made between red flowered and white flowered plants. What will be the phenotypic ratio in $F_2$ generation

A) $1:2:1$
B) $9:3:3:1$
C) $3:1$
D) $1:3$

132. When a heterozygous dominant is crossed with homozygous recessive than ratio of progeny will be

A) $1:2$
B) $2:1$
C) $3:1$
D) $1:1$

133. Mendel chose pea plants because

A) They were cheap
B) They were having seven pairs of contrasting characters
C) They were easily available
D) Of great economic importance

134. If a cross is made between $AA$ and $aa$, the nature of $F_1$ progeny will be

A) Genotypically $AA$, phenotypically $a$
B) Genotypically $Aa$, phyenotypically $a$
C) Genotypically $Aa$, phyenotypically $A$
D) Genotypically $aa$, phyenotypically $A$

135. The branch of botany dealing with heredity and variation is called

A) Geobotany
B) Sericulture
C) Genetics
D) Evolution

## Biology - ( Botany )

136. Read the following statements and choose the set of correct statements:
$(a)$ Euchromatin is loosely packed chromatin
$(b)$ Heterochromatin is transcriptionally active
$(c)$ Histone octomer is wrapped by negatively charged DNA in nucleosome
$(d)$ Histones are rich in lysine and arginine
$(e)$ A typical nucleosome contains $400\ bp$ of DNA helix
Choose the correct answer from the options given below:

A) $(a), (c), (d)$ Only
B) $(b), (e)$ Only
C) $(a), (c), (e)$ Only
D) $(b), (d), (e)$ Only

137. Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes?

A) $DNA$ dependent $DNA$ polymerase
B) $DNA$ dependent $RNA$ polymerase
C) $DNA$ Ligase
D) $DNase$

138. Matching sequence of $DNA$ between two evidences, one of the criminal with the suspect is known as:

A) $DNA\ fingerprinting$
B) $DNA\ amplification$
C) $Gene\ mapping$
D) $DNA\ resolution$

139. Given below are two statements:
Statement $I$: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.
Statement $II$: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.
In the light of the above statements, choose the correct answer from the options given below:

A) Statement $I$ is incorrect but Statement $II$ is true.
B) Both Statement $I$ and Statement $II$ are true.
C) Both Statement $I$ and Statement $II$ are false.
D) Statement $I$ is correct but Statement $II$ is false.

140. A bacterium grown over medium having radioactive Sulphur incorporates radioactivity in

A) $carbohydrates$
B) $proteins$
C) $DNA$
D) $RNA$

141. Which of the following statement is correct about $DNA$ polymerase?

A) $DNA$ polymerase can synthesize $mRNA$ in the $3' to 5'$ direction.
B) $DNA$ polymerase can synthesize $DNA$ in the $5' to 3'$ direction.
C) $DNA$ polymerase can synthesize $mRNA$ in the $5'$ to $3'$ direction.
D) $DNA$ polymerase can synthesize $DNA$ in the $3'$ to $5'$ direction.

142. Lactose operon produces enzymes

A) $b-$galactosidase, permease and glycogen synthetase.
B) $b-$galactosidase, permease and transacetylase.
C) permease, glycogen synthetase and transacetylase.

D) $b-$galactosidase, permease and phosphoglucose isomerase.

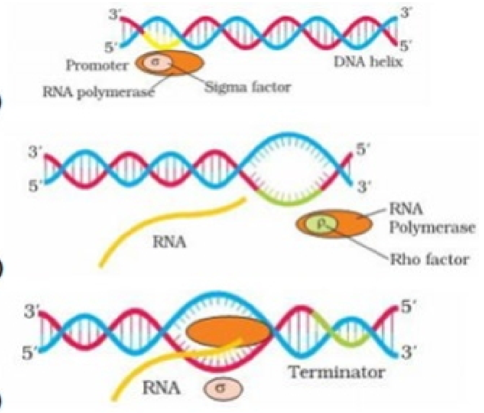
**143.** Choose the correct statement.

A) Transcription and translation occur in same compartment for prokaryotes
B) Monocistonic RNA - more than one structural genes under single promoter
C) Introns and exons both code for protein synthesis
D) In prokaryotes, splicing and tailing occurs before translation.

**144.** Codons of glycine are

A) $CCU, CCC, CCA, CCG$
B) $CGU, CGC, CGA, CGG$
C) $GGU, GGC, GGA, GGG$
D) $ACU, ACC, ACA, ACG$

**145.** Identify $a, b$ and $c$




(a)
(b)
(c)

A) $(a)$ Elongation, $(b)$ Termination, $(c)$ Initiation
B) $(a)$ Initiation, $(b)$ Termination, $(c)$ Elongation
C) $(a)$ Initiation, $(b)$ Elongation, $(c)$ Termination
D) $(a)$ Termination, $(b)$ Elongation, $(c)$ Initiation

**146.** In Meselson and Stahl's experiments, heavy $DNA$ was distinguished from normal $DNA$ by centrifugation in

A) $CsOH\ gradient$　B) $14\mathrm{NH4Cl}$
C) $15\mathrm{NH4Cl}$　D) $CsCl\ gradient$

**147.** Allelic sequence variations where more than one variant (allele) at a locus in a human population with a frequency greater than $0.01$ is refered to as

A) Incomplete dominance　B) Multiple allelism
C) $SNP$　D) $DNA$ polymorphism

**148.** Who proved that $DNA$ is basic genetic material?

A) Mendel　B) Watson
C) Boveri and Sutton　D) Hershey and Chase

**149.** Under which of the following conditions will there be no change in the reading frame of following $mRNA\ 5'AACAGCGGUGCUAUU3'$

A) Insertion of $G$ at $5^{\text{th}}$ position
B) Deletion of $G$ from $5^{\text{th}}$ position
C) Insertion of $\mathrm{A}$ at $\mathrm{G}$ at $4^{\text{th}}$ and $5^{\text{th}}$ positions respectively
D) Deletion of $GGU$ from $7^{\text{th}}, 8^{\text{th}}$ and $9^{\text{th}}$ positions

**150.** Assertion : " $DNA$ finger printing" has become a powerful tool to establish paternity and identity of criminals in rape and assault cases.
Reason : Trace evidences such as hairs, saliva and dried semen are adequate for $DNA$ analysis.

A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
C) If the Assertion is correct but Reason is incorrect.
D) If both the Assertion and Reason are incorrect.

**151.** Which of the following statements is correct?

A) Adenine does not pair with thymine
B) Adenine pairs with thymine through two $H-$ bonds
C) Adenine pairs with thymine through one $H-$ bond
D) Adenine pairs with thymine through three $H-$ bonds

**152.** $AGGTATCGCAT$ is a sequence from the coding strand of a gene. What will be the corresponding sequence of the transcribed $mRNA$?

A) $AGGUAUCGCAU$　B) $UCCAUAGCGUA$
C) $ACCUAUGCGAU$　D) $UGGTUTCGCAT$

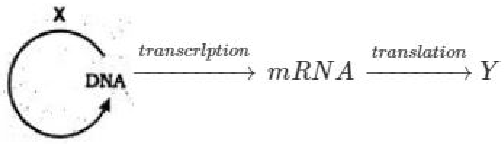
**153.** Which one of the following pairs of nitrogenous bases of nucleic acids, is wrongly matched with the category mentioned against it ?

A) Guanine, Adenine - Purines
B) Adenine, Thymine - Purines
C) Thymine, Uracil - Pyrimdines
D) Uracil, Cytosine - Pyrimidines

**154.** Identify the correct order of organisation ofgenetic material from largest to smallest.

A) Genome, chromosome, gene, nucleotide
B) Chromosome, genome, nucleotide, gene
C) Chromosome, gene, genome, nucleotide
D) Genome, chromosome, nucleotide, gene

**155.** By using diagram, select the correct option for $X$ and $Y$




A) Replication　Protein
B) Cell division　Protein
C) Replication　Amino acid
D) $t-RNA$　$r-RNA$

**156.** The association of histone $H_1$ with a nucleosome indicates that

A) $DNA$ replication is occurring
B) the $DNA$ is condensed into a chromatin fibre
C) the $DNA$ double helix is exposed
D) transcription is occurring.

**157.** Select the correct option.Direction of Direction of reading of $RNA$ synthesis the template $DNA$ strand

| Direction of $RNA$ synthesis | Direction of reading of the template $DNA$ |
|---|---|
| $(a)$ $5' - 3'$ | $3' - 5'$ |
| $(b)$ $3' - 5'$ | $5' - 3'$ |
| $(c)$ $5' - 3'$ | $5' - 3'$ |
| $(d)$ $3' - 5'$ | $3' - 5'$ |

A) $(a)$ B) $(b)$ C) $(c)$ D) $(d)$

158. Which of following is not salient feature of human genome ?

A) Human genome contains $3164.7$ million nucleotide base.
B) The function are unknown for over $50$ percent of the discovered genes.
C) More than $2$ percent of the genome codes for proteins.
D) Chromosomes $1$ has most genes and they has the fewest.

159. Formation of $RNA$ strand on template $DNA$ is

A) Replication . B) Transcription
C) Protein synthesis D) Synthesis of primer

160. $RNA$ as genetic material in ............

A) $TMV$ virus B) Cyano bacteria
C) Clado phora D) Sieve cells

161. The modern concept of gene is

A) A segment of $DNA$ capable of crossing over
B) A functional unit of $DNA$
C) A segment of $DNA$
D) A segment of chromosome

162. Adenine is $30\%$ then what would be the $\%$ of guanine

A) $10\%$ B) $20\%$ C) $30\%$ D) $40\%$

163. $DNA$ is double helix and

A) Complementary and parallel
B) Complementary and antiparallel
C) Without super coils
D) Always circular

164. Transcription of $DNA$ is aided by

A) $RNA$ polymerase B) $DNA$ polymerase
C) Exonuclease D) Recombinase

165. On which part of $tRNA$, amino acid gets linked

A) Anticodon loop B) Variable loop
C) $3$' end D) $5$' end

166. $DNA$ strand is directly involved in the synthesis of all the following except

A) $tRNA$ molecule
B) $mRNA$ molecule
C) Another $DNA$ strand
D) Protein synthesis

167. Repressor protein is formed from

A) Repressor gene B) Structural gene
C) Operator gene D) Regulatory gene

168. Property of a codon for always coding a specific amino acid is called

A) Degenerate B) Non-ambiguous
C) Non-overlapping D) None of the above

169. $DNA$ is not directly involved with the synthesis of the following

A) $m-RNA$ B) $r-RNA$ C) $t-RNA$ D) Protein

170. Synthesis of any protein in a cell determines

A) Types of ribosomes
B) Mitochondria
C) Sequence of nucleotides in $DNA$
D) Suger and phosphate of $DNA$

171. Which one of the following codons codes for the same information as $UGC$

A) $UGU$ B) $UGA$ C) $UAG$ D) $UGG$

172. The arrangement of three bases in the genetic code signifies a specific

A) Protein B) Amino acid
C) Plasmid D) Nucleic acid

173. In Operon concept, the regulator gene regulates chemical reactions in the cell by

A) Inactivating enzymes in the reaction
B) Inhibiting transcription of $mRNA$
C) Inhibiting migration of $mRNA$ into cytoplasm
D) Inhibiting the substrate in the reaction

174. The double helix model of Watson and Crick is known as

A) $C-DNA$ B) $B-DNA$
C) $Z-DNA$ D) $D-DNA$

175. Eukaryotic $RNA$ polymerase $III$ catalyse the synthesis of

A) $m\ RNA$ B) $t\ RNA$
C) $18.5\ rRNA$ D) Introns

176. Phosphorus is present in

A) Protein B) $DNA$
C) $RNA$ D) Both $DNA$ and $RNA$

177. Bacteria were grown in a medium containing heavy isotope of nitrogen ($N^{15}$) for many generations and all their $DNA$ contained many heavy nitrogen only. A bacterium of this type was transferred to normal medium and allowed to duplicate. After two divisions of heavy $DNA$ is likely to be that

A) Only one daughter cell will have heavy $DNA$
B) Two daughter cells have normal $DNA$ and other two have both normal and heavy $DNA$
C) All daughter cells have heavy $DNA$
D) Half daughter cells have heavy $DNA$ and other half have normal $DNA$

178. The length of a helix of $DNA$ is

A) $10Å$ B) $20Å$
C) $30Å$ D) $34Å$

179. The element absent in $RNA$ is

A) Nitrogen B) Sulpher C) Oxygen D) Hydrogen

180. Isolation and purification of specific $DNA$ segment from a living organism was achieved by

A) Crick B) Nirenberg
C) Khorana D) Beckwith and his colleagues